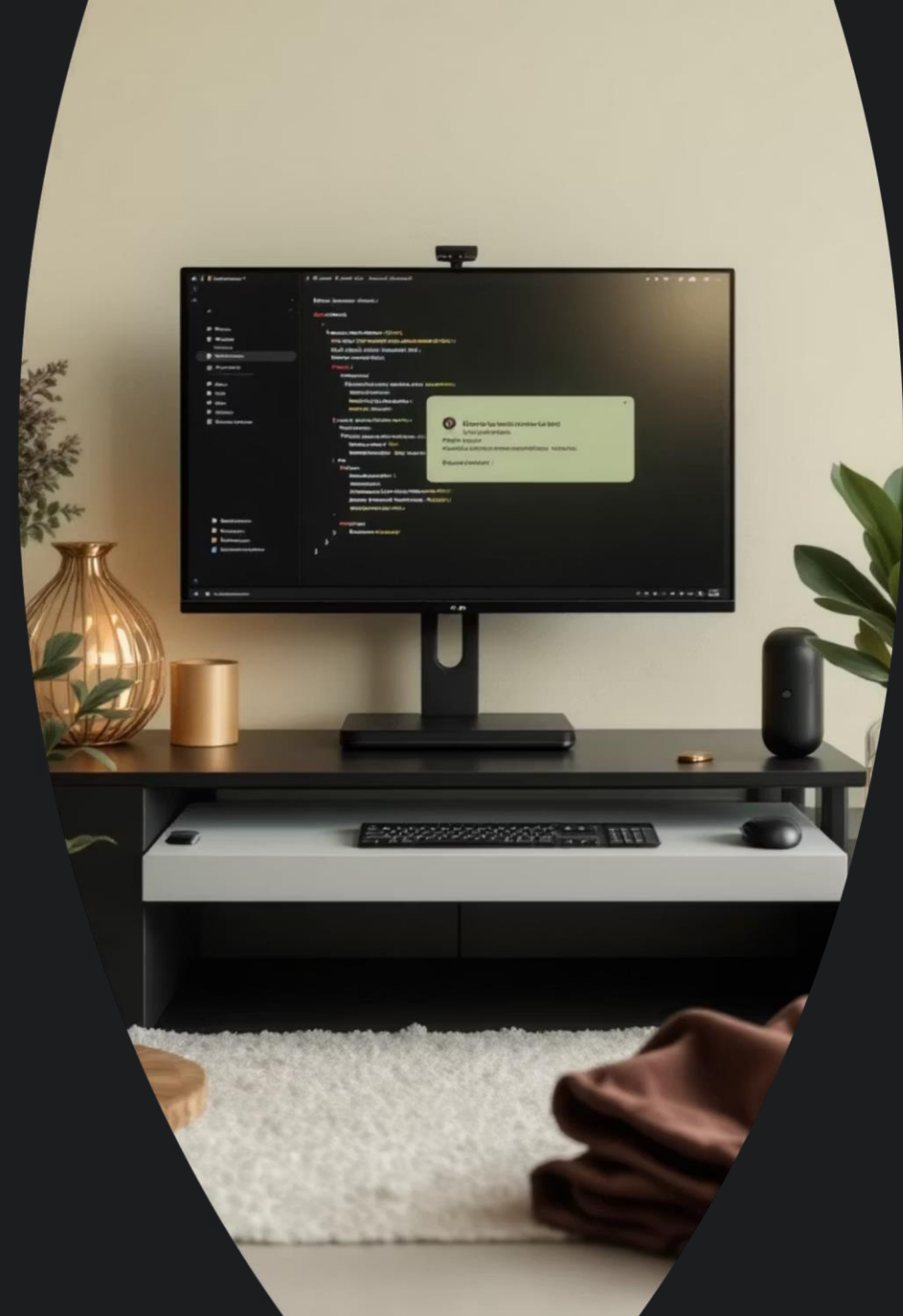




Research Paper Answer Bot

Intelligent RAG System for Generative AI Research Papers

Analytics Vidya Capstone Project

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Project Objectives

01

Data Foundation

Load and index seminal Generative AI research papers into a vector database system

03

Retrieval Optimization

Test various retrieval strategies from simple cosine similarity to hybrid search search with rerankers

02

Embedding Exploration

Experiment with multiple embedding models from HuggingFace and commercial providers like OpenAI

04

LLM Integration

Connect vector database to large language models for intelligent response response generation

RAG System

1

Document Processing

PDF extraction, chunking strategies, and preprocessing optimization

2

Vector Database

Embedding storage with multiple model support and efficient indexing

3

Retrieval Engine

Hybrid search combining semantic similarity with keyword matching

4

Response Generation

LLM integration with context-aware answer synthesis

Technical Implementation

1

Embedding Models Comparison

Tested sentence-transformers, OpenAI text-embedding-ada-002, and domain-specific models for optimal semantic understanding

2

Retrieval Strategy Evolution

Progressed from simple cosine similarity to hybrid search incorporating BM25 rerankers for improved relevance

3

Vector Database Optimization

Implemented efficient indexing with FAISS, supporting multiple embedding dimensions and fast similarity search

4

LLM Response Synthesis

Integrated GPT-4.1 with context-aware prompting and citation tracking for accurate, traceable answers

Implementation Challenges & Solutions

Document Chunking Dilemma

Research papers contain complex cross-references.

1

Solution: Implemented semantic chunking that preserves context boundaries.

Embedding Model & Evaluation

Different models showed varying performance on technical content.

3

Solution: Created evaluation framework comparing retrieval accuracy.

LLM Compatibility

Different behaviors across Open Source & Commercial LLMs

2

Solution: Built a unified approach ensuring consistency

User Experience

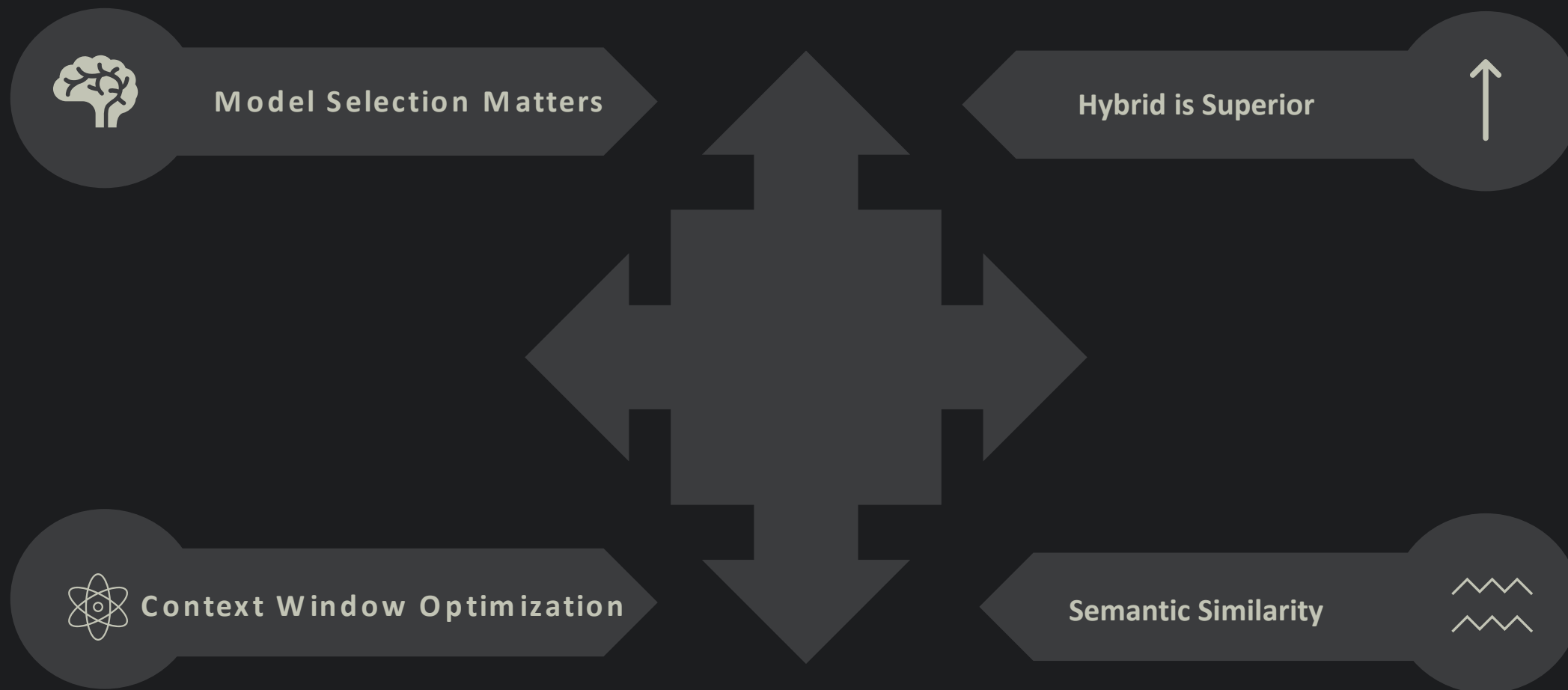
Ensuring smooth end-to-end usability

4

Solution: Designed Streamlit UI to display every step seamlessly



Key Learnings



UI Snippets (Stretch Goal)

Configuration

Select Embedding Model ?

openai

BAAI/bge-small-en

openai

500

2000

Chunk Overlap

200

50

500

Number of Documents to Retrieve

5

1

10

System Information

Current Configuration:

- Embedding Model: openai
- Chunk Size: 1000
- Chunk Overlap: 200
- Retrieval K: 5



Research Paper Answer Bot

Powered by RAG (Retrieval-Augmented Generation)



Upload Research Papers

Choose PDF files



Drag and drop files here
Limit 200MB per file • PDF

Browse files



Ask Questions

Enter your question:

Ask anything about the research papers...

Get Answer

Answer

The main components of a Transformer architecture are:

1. **Encoder and Decoder Stacks:** Both the encoder and decoder are composed of a stack of $N = 6$ identical layers.
2. **Multi-Head Self-Attention Mechanism:** Each layer in the encoder and decoder has a multi-head self-attention mechanism as its first sub-layer.
3. **Position-Wise Fully Connected Feed-Forward Network:** The second sub-layer in each layer is a position-wise fully connected feed-forward network.
4. **Residual Connections and Layer Normalization:** Residual connections are employed around each of the two sub-layers, followed by layer normalization.
5. **Embedding Layers:** These layers produce outputs of dimension ($d_{\text{model}} = 512$) to facilitate residual connections.

These components work together to allow the Transformer to compute representations of its input and output sequences without using sequence-aligned RNNs or convolution, enabling significant parallelization and improved performance on tasks like machine translation.

(Source: Context provided)



Sources

Source 1: attention_paper.pdf (Page 2)

Source 2: attention_paper.pdf (Page 1)

Source 3: attention_paper.pdf (Page 0)

Deploy



Deploy

Or choose a sample:

What are the main components of a transfo... ▼

What are the main components of a transfo...

How does attention mechanism work in neural ...

What are the advantages of using pre-trained la...

Explain the concept of transfer learning in NLP.

What are the challenges in training large langu...

Q&A

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